

DataAnalyticsInternship

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Domain: Data Analytics

Project: Data-Driven Insights into OLA Ride Bookings using Power BI Dashboard

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PHASE1—BLUEPRINTDOCUMENT

ProjectTitle

Data-Driven Insights into OLA Ride Bookings using Power BI Dashboard

ProjectType

GroupProject

1. ProblemStatement

Ride-hailingplatformslikeOLAgeneratelargevolumesofbookingandridedata.However, without proper analysis, it becomes difficult to:

- Understandbookingtrends
- Identifycancellationpatterns
- Evaluatevehicleperformance
- Trackrevenueandrideefficiency

Traditionalreportsfailtoprovideactionableinsights,makingitchallengingforbusiness stakeholders to optimize operations and improve customer satisfaction.

2. DatasetDetails

DatasetName: OLABookingsDataset

SourceFilesUsed:

- bookings.csv(Primarydataset)
- ola.sql(Databasequeries&views)
- ola.pbix(PowerBI dashboard)

Type:Structureddata(CSV+SQLdatabase)

Description:

Thedatasetcontainsridebookinginformation includingridestatus,vehicletype,distance, and revenue details.

KeyColumns:

- BookingID
- BookingStatus(Success/Cancelled)
- CustomerID
- Vehicle Type
- RideDistance
- BookingValue(Fare)
- Pickup&Drop Location
- BookingDate/Time

3. Hypothesis

- Higher cancellations reduce overall revenue
- Certain vehicle types generate more revenue than others
- Ride distance impacts booking values significantly
- Peak hours contribute to higher booking volumes
- Customer cancellations are more frequent than driver cancellations

4. Solution Approach

The project follows a structured analytics pipeline:

Step 1: Data Preparation

- Load CSV into SQL database
- Clean missing and inconsistent values
- Create derived columns if needed

Step 2: SQL Analysis (ola.sql)

- Create views such as: Successful bookings, Cancelled rides, Average ride distance per vehicle
- Perform aggregation queries

Step 3: Data Analysis (Python – optional)

- Exploratory Data Analysis (EDA)

- Trendidentification
- Patternrecognition

Step4:PowerBIDashboard(ola.pbix)

- Importcleaneddataset
- Buildinteractivevisuals
- Applyfiltersandslicers

Step5:InsightsGeneration

- Identifybusinesstrends
- Highlightperformanceissues
- Providerecommendation

5. AnalysisPlanwithExpectedVisualizations

OverallPerformance

→KPICards

- TotalBookings
- TotalRevenue
- TotalDistance
- SuccessRate

BookingStatusAnalysis

→ Pie/Donut Chart

- SuccessvsCancelledrides

VehiclePerformance

→BarChart

- RevenuebyVehicleType
- AverageRideDistance

Time-BasedAnalysis

→LineChart

- Bookingsvertime

- Peakhourdemand

CancellationAnalysis

→BarChart

- CancellationbyCustomervs Driver

DistancevsRevenue

→Scatter Plot

- Relationshipbetweenridedistance andfare

TopInsights

→Table /BarChart

- Topperformingvehicle
- Mostfrequentroutes

6. Tech Stack

- SQL(MySQL/PostgreSQL)–Dataquerying(ola.sql)
- Python(JupyterNotebook) –OptionalEDA
- Libraries:Pandas,NumPy
- Visualization:Matplotlib,Seaborn
- DashboardTool:PowerBI(ola.pbix)

7. Success Criteria

- Dashboardclearlypresentskeyridemetrics
- Visualizationsareinteractiveandinsightful
- Businessinsightsareeasytointerpret
- Trendslike cancellationsandpeakdemandare identified
- Decision-makingissupportedthroughdata